Financial Management

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Module 2: Part 2
Classification of Risk & Leverage Analysis

Classification of Risk

- Business Risk Vs. Financial Risk
- Market Risk, Credit Risk and Operational Risk

Business Risk

- Anything that threatens a company's ability to meet its target or achieve its financial goals is called business risk.
- It is related to the overall operations of the business
- Business risk is influenced by a number of different factors including:
  - Consumer preferences, demand, and sales volumes
  - Per-unit price and input costs
  - Competition
  - The overall economic climate
  - Government regulations

Financial Risk

- Risk of loss of capital
- Financial risk of a company is possibility of loosing future cash inflows to meet company's obligations.
- Financial risk is a term that can apply to businesses, government entities, the financial market as a whole, and the individual.
- This risk is the danger or possibility that shareholders, investors, or other financial stakeholders will lose money.
- Financial risks include credit risk, liquidity risk, and operational risk.

Market Risk

- Market risk is the possibility of an investor experiencing losses due to factors that affect the overall performance of the financial markets in which he or she is involved.
- Also known as ***"Systematic Risk"***
- Market risk *cannot* be eliminated through diversification, though it can be hedged against in other ways.
- Market risk of a stock represents that portion of its risk which is attributable to economy-wide factors like the growth rate of GNP, the level of government spending, money supply, interest rate structure, and inflation rate.

Credit Risk

- Credit risk is the possibility of a loss resulting from a borrower's failure to repay a loan or meet contractual obligations.
- Also known as risk of defaulting.
- Proper assessment of debtors/borrowers can help in management of such risk.

Operational Risk

- Pertaining to business' day-to-day activities
- Operational risk can also be classified as a variety of ***unsystematic risk***, which is ***unique*** to a specific company or industry.
- The unique risk of a security represents that portion of its total risk which stems from firm-specific factors like development of a new product, a labour strike, machine break-down, or an entry of new competitors.
- It is diversifiable.
- The unique risk of investment can be diversified by creating diversified portfolio.

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2.4: BEP-Leverage Analysis

Break-even Analysis

- Break-Even Analysis – A technique for studying the relationship among fixed costs, variable costs, sales volume, and profits. Also called cost/volume/profit analysis (C/V/P) analysis.
- Break-Even Point – The sales volume required so that total revenues and total costs are equal; may be in units or in sales dollars
- When studying operating leverage, “profits” refers to operating profits before taxes (i.e., EBIT) and excludes debt interest and dividend payments.
- BEP when EBIT = 0

Leverage: Operating Leverage

- **Operating Leverage:**
  - Arises from the existence of fixed operating expenses
  - When firm has fixed operating expenses, 1 % change in sales leads to more than 1% change in EBIT

Degree of Operating Leverage (DOL) = $\frac{Q\ (P-V)}{Q(P-V)-F}$

$$=\frac{Contribution}{Earnings\ before\ Interest\ and\ Taxes(EBIT)}$$

Contribution = selling price – VC

EBIT = contribution – fixed cost

Operating Leverage

Example: Finex Ltd.'s Price is Rs. 1500, Variable cost is Rs. 1000, and Fixed cost is Rs. 200000. Calculate OL at the level of output Q = 500 units and Q = 600 units.

Degree of Operating Leverage (DOL) = $\frac{Q\ (P-V)}{Q(P-V)-F}$

DOL (Q = 500) $= \frac{500\ (1500-1000)}{500(1500-1000)-200000} = 5.0$

DOL (Q = 600) $= \frac{600\ (1500-1000)}{600(1500-1000)-200000} = 3.0$

Financial Leverage

- Emanates from the existence of fixed interest expenses
- When the firm has fixed interest expenses, 1 % change EBIT leads to more than 1% change in PBT (or Net Income).

Degree of Financial Leverage (DFL) $= \frac{EBIT}{EBIT - I}$

$$= \frac{EBIT}{Earning\ Before\ Tax\ (EBT)}$$

Financial Leverage

Example: Finex Ltd. Is having currently an interest expense of Rs. 30000. Then the FL at EBIT = Rs. 50000 and EBIT = Rs. 60000 will be

Degree of Financial Leverage (FL) $= = \frac{EBIT}{EBIT - I}$

DFL at (EBIT = Rs.50000) $= \frac{50000}{50000 - 30000} = 2.5$

DFL (EBIT = Rs. 60000) $= \frac{60000}{60000 - 30000} = 2.0$

Total or Combined Leverage

- Arises from the existence of fixed operating cost and interest expenses.
- 1% change in revenues leads to more that 1% change in PBT.

Degree of Total Leverage (DTL) $= \frac{Contribution}{EBT}$

Degree of Total Leverage (DTL) = DOL X DFL

= 5 X 2.5 = 12.5

Summary

Operating Leverage	Financial Leverage	Business Risk	Financial Risk
$\frac{Contribution}{EBIT}$	$\frac{EBIT}{PBT}$	Variability in the firm's expected EBIT.	Additional variability in net income to common shareholders.
Increases with higher fixed cost	Increases with higher debt	Increases with high OL.	Increases with high FL.
If a firm already has high business risk, then it may be advisable to use less debt to lower the financial risk. If a firm has less business risk, then it may be able to afford higher financial risk.			

Combination of OL & FL and their combined effect

Operating Leverage	Financial Leverage	Combined Leverage
High	High	Very dangerous policy, must be avoided
Low	Low	Very cautious policy and not assuming risk
High	Low	Adverse effects of OL and are taken care by having low FL
Low	High	An ideal situation. The company can follow aggressive debt policy

Practice Problems

VST corporation has sales of Rs.40 lakhs, variable cost 70 per cent of the sales and fixed cost is Rs.8 lakhs. The firm has raised Rs.20 lakhs funds by issue of debentures at the rate of 10 per cent. Compute DOL, DFL, and DCL.

Practice Problems

Ex.1: The following figures relate to two companies:

Particulars	X Ltd. (Rs in Lakh)	Y Ltd. (Rs. In Lakh)
Sales	250	500
Variable Costs	100	150
Contribution	150	350
Fixed Costs	75	200
EBIT	75	150
Interest	25	50
PBT	50	100

You are required to:

1. Calculate DOL, DFL and DTL for the two companies
2. Comment on the relative risk position of them

Practice Problems

Ex.2:

(i) Find the operating leverage from the following data:

Sales	Rs. 100000
variable Costs	60%
Fixed Costs	Rs. 24000

(ii) Find the financial leverage from the following data:

Net Worth	Rs. 5000000
Debt/Equity	3/1
Interest Rate	12%
Operating Profit	Rs. 4000000

Practice Problems

Ex. 3: Calculate the operating leverage, financial leverage and combined leverage from the following data under Situation I and II and Financial Plan A and B.

Installed Capacity 4000 units

Actual Production and Sales 75% of the Capacity

Selling Price Rs. 30 per unit

Variable cost Rs. 15 per unit

Fixed Cost:

Under Situation I Rs. 15000

Under Situation II Rs. 20000

Capital Structure:

Particular	Plan A	Plan B
Equity	Rs. 10000	Rs. 15000
Debt (Rate of Interest @ 12%)	Rs. 10000	Rs. 5000

THANK YOU!!!